

CO 3 - ASSESSMENT TOOL 2: Data Interpretation

SIMATS ENGINEERING

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Statistics Exercises — Sl. No 16–20

16. Continuous Distribution — Area Interpretation

Question

A continuous random variable X is normally distributed with $\mu = 100$ and $\sigma = 15$.

(a) What proportion of values lie between 85 and 130?

(b) A value of 145 is observed — how many standard deviations away is it, and would you flag it as a potential outlier under the empirical rule?

Solution

(a) Answer:

For $X = 85$:

$$Z = (85 - 100) / 15 = -1$$

For $X = 130$:

$$Z = (130 - 100) / 15 = 2$$

From the standard normal table:

$$P(Z < 2) = 0.9772$$

$$P(Z < -1) = 0.1587$$

Therefore:

$$P(85 < X < 130) = 0.9772 - 0.1587 = 0.8185$$

Answer = 81.85%

(b) Answer:

For $X = 145$:

$$Z = (145 - 100) / 15 = 3$$

Therefore, 145 is 3 standard deviations above the mean.

Yes, it can be flagged as a potential outlier under the empirical rule.

Explanation

The z-score tells us how many standard deviations a value is away from the mean. Approximately 81.85% of values lie between 85 and 130. The value 145 is 3 standard deviations above the mean, so it is unusually far from the average.

17. Sampling Bias vs Sampling Error

Question

A survey estimating average household expenditure samples only online panel respondents. The sample mean is ₹18,000 while the known population mean is ₹14,500 from census data.

- (a) Is this discrepancy better explained by sampling error or sampling bias?
- (b) Would increasing the sample size fix this problem? Justify.

Solution

(a) Answer:

The discrepancy is better explained by sampling bias.

(b) Answer:

No, increasing the sample size alone will not fix the problem. A representative sampling method is required.

Explanation

The survey includes only online panel respondents, so some groups in the population may be excluded or underrepresented. This creates sampling bias. Increasing the sample size reduces random sampling error, but it does not remove systematic bias. Therefore, even a very large biased sample can produce an inaccurate estimate.

18. Covariance Sign vs Correlation Strength

Question

Two variable pairs both show positive covariance:

- Pair 1: Covariance = 850
- Pair 2: Covariance = 12

Explain why you cannot conclude that Pair 1 has a "stronger relationship" than Pair 2 without additional information, and state what that missing information is.

Solution

Answer:

We cannot conclude that Pair 1 has a stronger relationship simply because its covariance is 850 compared with 12.

The missing information is the standard deviations of the variables, or preferably the correlation coefficient.

$$r = \text{Cov}(X,Y) / (\sigma X \times \sigma Y)$$

Explanation

Covariance depends on the scale and units of the variables. Therefore, a larger covariance does not necessarily mean a stronger relationship. Correlation standardizes covariance and allows relationships to be compared properly. The correlation coefficient ranges from -1 to 1. Hence, correlation is more suitable for comparing relationship strength.

19. Estimation — Confidence Interval Reasoning

Question

A sample of $n = 100$ items has mean $= 250$ and $SD = 40$. Construct a 95% confidence interval for the population mean. Then explain, in interpretation terms, what this interval does and does NOT tell you about the true population mean.

Solution

Answer:

Given: $n = 100$, $\bar{x} = 250$, $s = 40$

For a 95% confidence level: $Z = 1.96$

Standard error: $SE = 40 / \sqrt{100} = 4$

Margin of error: $1.96 \times 4 = 7.84$

Therefore: $CI = 250 \pm 7.84$

Lower limit $= 250 - 7.84 = 242.16$

Upper limit $= 250 + 7.84 = 257.84$

95% Confidence Interval $= (242.16, 257.84)$

Interpretation Answer:

The interval gives a plausible range of 242.16 to 257.84 for the true population mean.

It does not mean that there is a 95% probability that the fixed population mean is inside this particular interval.

Explanation

A confidence interval shows the uncertainty associated with estimating a population mean from a sample. A 95% confidence procedure is designed so that, over repeated samples, approximately 95% of the resulting intervals would contain the true population mean.

20. Integrated Case Study

Question

A retail company's daily sales data in ₹'000 for 15 days is:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

- (a) Identify the outlier and justify using both the IQR method and z-score method.
- (b) Compute mean and median with and without the outlier — compare the sensitivity of each.
- (c) Compute skewness with the outlier included — classify the distribution shape.
- (d) Recommend whether the company should remove, cap, or investigate the outlier before reporting monthly performance.

Solution

(a) Answer:

Arrange the data:

117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500

$$Q1 = 120$$

$$Q3 = 128$$

$$IQR = 128 - 120 = 8$$

$$\text{Upper boundary} = Q3 + 1.5(IQR) = 128 + 1.5(8) = 140$$

Since $500 > 140$, 500 is an outlier by the IQR method.

For the z-score:

$$\text{Mean} = 148.47$$

$$SD \approx 94.03$$

$$Z = (500 - 148.47) / 94.03 \approx 3.74$$

Since $Z > 3$, 500 is also an outlier by the z-score method.

(b) Answer:

With the outlier:

$$\text{Mean} = 148.47$$

$$\text{Median} = 124$$

Without the outlier:

$$\text{Mean} = 123.36$$

$$\text{Median} = 123.5$$

Mean changes considerably from ₹123.36k to ₹148.47k, while the median changes only slightly from ₹123.5k to ₹124k.

(c) Answer:

Skewness with the outlier included is approximately 3.86.

Therefore, the distribution is strongly positively skewed (right-skewed).

(d) Answer:

The company should investigate the outlier first rather than immediately removing or capping it.

- IQR method identifies 500 as an outlier.
- Z-score is approximately 3.74.
- Mean changes considerably from ₹123.36k to ₹148.47k.
- Median changes only slightly from ₹123.5k to ₹124k.
- Skewness is approximately 3.86, showing strong right skewness.

If 500 represents a genuine large sale, it should be retained. If it is a data-entry error or incorrect transaction, it should be corrected or removed.

Explanation

The value 500 is clearly unusual and has a major effect on the mean and distribution shape. However, an outlier should not automatically be deleted. The company should first determine whether the value represents a real business event or a data error. This makes investigation the most appropriate first step.